Ed-Tech Learning Analytics & Student Performance Analysis

Abstract

Educational institutions increasingly rely on data-driven approaches to understand student performance and improve learning outcomes. This project analyzes student demographic, behavioral, and engagement data to identify key factors influencing academic performance and to detect students at risk of underperformance. Using SQL, Python-based analytics, machine learning, and Power BI dashboards, the project delivers actionable insights to support early intervention and informed academic decision-making.

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1. Introduction

Student academic performance is influenced by a combination of behavioral, environmental, and demographic factors. Traditional evaluation methods often focus only on final grades, overlooking early warning indicators such as attendance patterns, study habits, and engagement levels.

This project aims to bridge that gap by building an end-to-end analytics solution that:

- Examines academic performance drivers
- Predicts student outcomes using machine learning
- Visualizes insights through interactive dashboards
- Enables early identification of at-risk students

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2. Objectives

The primary objectives of this project are:

1. To analyze overall student academic performance using GPA and grade classifications
2. To evaluate the impact of behavioral factors such as study time, attendance, tutoring, and parental support
3. To identify the most influential predictors of student performance using machine learning
4. To classify students into risk levels for proactive academic intervention
5. To design an interactive Power BI dashboard for stakeholders and decision-makers

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3. Dataset Description

The dataset contains student-level academic and engagement data with the following attributes:

3.1 Demographic Attributes

- Student ID
- Age
- Gender
- Ethnicity
- Parental Education

3.2 Engagement & Behavioral Attributes

- Weekly Study Time
- Absences
- Tutoring
- Parental Support
- Extracurricular Activities
- Sports, Music, and Volunteering Participation

3.3 Academic Performance Attributes

- GPA
- Grade Classification

Encoded categorical variables were validated and transformed into human-readable labels during data preprocessing to ensure accurate interpretation and reporting.

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4. Tools & Technologies

Category	**Tools**
Data Extraction	SQL
Data Analysis & Modelling	Python (Pandas, NumPy, Matplotlib, Scikit-learn)
Data Transformation	Power Query
Visualization & Reporting	Power BI
Machine Learning	Random Forest Regression

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5. Data Preparation & Feature Engineering

Data preparation involved multiple stages:

- Cleaning missing and inconsistent values
- Standardizing categorical variables
- Validating grade encodings against GPA distributions
- Creating engineered features such as:
    - Average daily study hours
    - Study efficiency
    - Risk score and risk level (Low, Medium, High)

These steps ensured data integrity and improved the reliability of downstream analysis.

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6. Exploratory Data Analysis (EDA)

Exploratory analysis revealed several important patterns:

- GPA distribution showed clear segmentation across student groups
- Higher study time was generally associated with improved GPA
- Absences had a strong negative relationship with academic performance
- Parental support and tutoring showed measurable positive effects
- Demographic factors had relatively lower influence compared to behavioral attributes

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7. Machine Learning Model

Train Random Forest

```
model = RandomForestRegressor(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

Evaluate the model

```
print("R2 Score:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
```

```
R2 Score: 0.9255568553946584
MAE: 0.18569821428571423
```

Feature Importance

```
Importance = pd.Series(model.feature_importances_, index=X.columns)
Importance.sort_values(ascending=False).head(10)
```

```
absences                0.868703
parental_support        0.029181
study_time_weekly       0.026975
avg_daily_study_hours   0.025970
tutoring                0.009976
parental_education      0.007099
age                     0.006428
ethnicity               0.006372
sports                  0.004834
extracurricular         0.004212
dtype: float64
```

7.1 Model Selection

A Random Forest Regressor was selected due to its ability to capture nonlinear relationships and handle mixed data types effectively.

7.2 Target Variable

- GPA

7.3 Model Performance

- R^2 Score: 0.93
- Mean Absolute Error (MAE): 0.19

The model demonstrated strong predictive performance while avoiding data leakage by excluding derived performance labels from the feature set.

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8. Feature Importance Analysis

Feature importance results indicated:

1. Absences as the dominant predictor of GPA
2. Parental support as a key positive influence
3. Study time (weekly and daily) as a consistent contributor
4. Tutoring as an effective academic intervention
5. Demographic attributes as secondary factors

This confirms that **student behavior and engagement are more influential than background characteristics**.

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9. Dashboard Design & Visualization

An interactive Power BI dashboard was developed and structured into three analytical layers.

9.1 Page 1 – Student Performance Overview

Purpose: Provide a high-level summary of academic outcomes.

Visuals Included:

- KPI cards for average GPA, study time, absences, and high-risk student count
- Grade distribution of students
- Performance group segmentation
- Average GPA by gender
- Matrix analysis of GPA by gender and parental support

Figure 1: Student Performance Overview Dashboard

gender	0	1	2	3	4	Total
0	1.78	1.68	1.99	2.20	2.40	2.00
1	1.20	1.76	1.99	1.98	2.26	1.91
Total	1.47	1.72	1.99	2.08	2.32	1.95

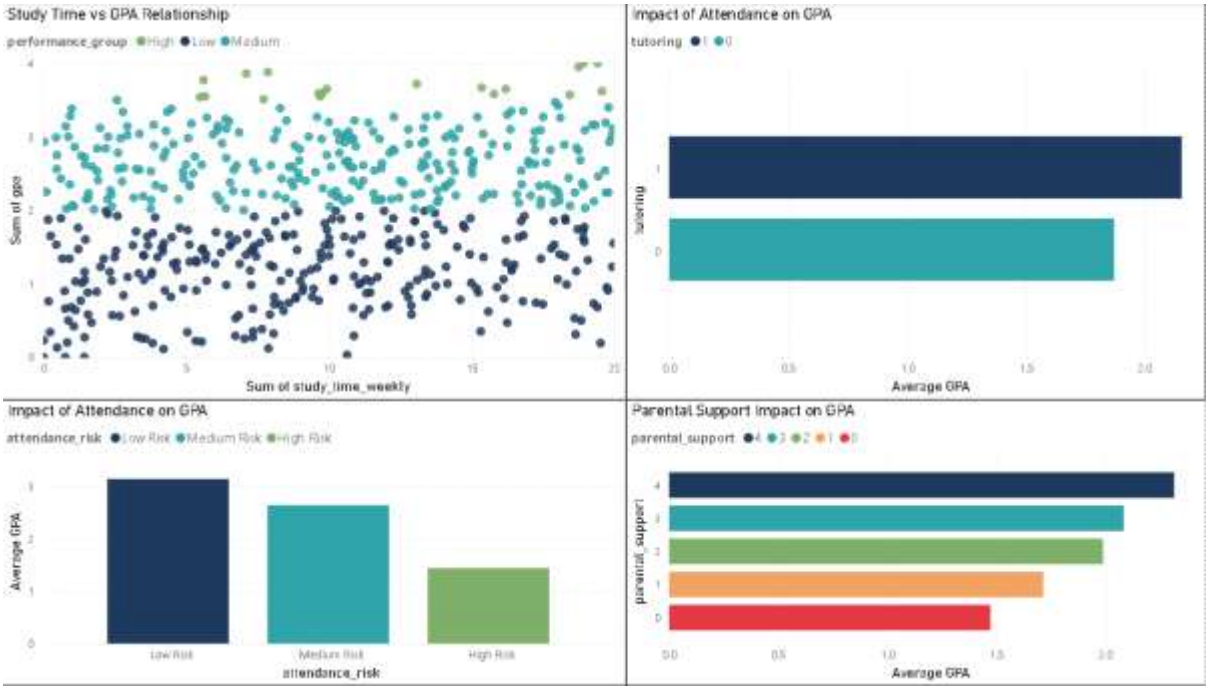
9.2 Page 2 – Engagement & Behavioral Insights

Purpose: Explain the behavioral drivers of academic performance.

Visuals Included:

- Study time vs GPA scatter plot
- Attendance impact on GPA
- Effect of tutoring on academic performance
- Parental support impact on GPA

Figure 2: Engagement & Behavioral Analysis Dashboard



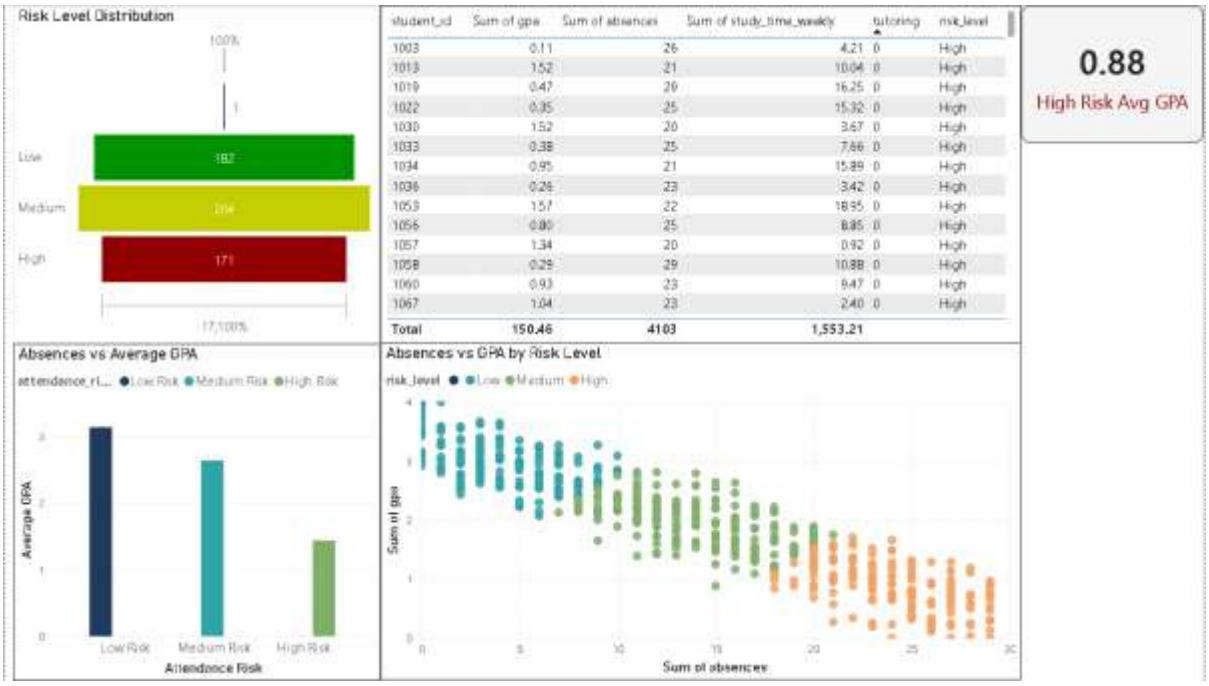
9.3 Page 3 – Student Risk Analysis

Purpose: Identify and prioritize students requiring intervention.

Visuals Included:

- Risk level distribution funnel
- Absences vs GPA analysis
- At-risk student identification table
- Absences vs GPA scatter plot by risk level

Figure 3: Student Risk Analysis Dashboard



Interactive tooltips were implemented across visuals to provide contextual explanations without cluttering the interface.

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10. Key Insights

- Attendance is the most critical factor influencing student performance
- Consistent study habits significantly improve academic outcomes
- Parental support and tutoring positively affect GPA
- Extracurricular activities have minimal direct academic impact
- Early identification of high-risk students enables targeted intervention

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11. Recommendations

Based on the analysis, the following recommendations are proposed:

1. Implement attendance monitoring and early-warning systems
2. Encourage structured study schedules and time management practices
3. Promote parental engagement initiatives
4. Provide targeted tutoring for high-risk students
5. Use analytics dashboards for continuous academic performance monitoring

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12. Conclusion

This project demonstrates the effectiveness of combining data analytics, machine learning, and visualization to understand student performance and identify academic risk. By translating analytical insights into an interactive dashboard, the solution supports informed decision-making and proactive academic interventions.

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13. Future Scope

- Incorporation of time-series data for longitudinal analysis
- Expansion to classification-based risk prediction models
- Integration with real-time academic data pipelines
- Deployment of dashboards for institutional use